Changes Since Milestone 2

Rahul Mehta, Ishaan Mohan, Zhengyu Chris Liu, Timothy Kim, Chau M Duong
ISYE 4031: Group 1 October 30, 2026

- **Data and evaluation**
  - **Method:** Reconstructed Census data; nested temporal validation; paired block-bootstrap comparisons.
  - **Result:** 706 forecasts at 10 dates (635 historical + 71 already in the original prospective check). No additional Census release beyond that check; original baseline reproduced.
- **Industry-specific models**
  - **Method:** Compared separate industry regressions and partial pooling with pooled OLS.
  - **Result:** Industry-model MAE: **3.04 percentage points**, versus 3.30 for pooled OLS and 3.39 for the recent average; 10.3% improvement over the average.
- **Nonlinear subset selection**
  - **Method:** Searched 320 hierarchical predictor-block combinations, including squares and interactions.
  - **Result:** Expected use squared selected at 6/10 dates; current-by-expected use at 4/10. Current use squared and time squared never selected. MAE: **3.32**; no improvement over baseline.
- **Advanced models and added predictors**
  - **Method:** Tested ridge, elastic net, and gradient boosting with lags, rolling averages, aggregate rates, interactions, and panel-cycle terms.
  - **Result:** MAE: elastic net **3.07**, boosting **3.12**, ridge **3.13**, full OLS **5.68**. Removing history worsened ridge to 3.70; cycle terms added little.
- **Industry growth and ranking uncertainty**
  - **Method:** Estimated industry trends with group intercepts; used block-bootstrap intervals and ranks.
  - **Result:** Leading growth: finance 1.28, Information 1.22, professional services 1.08 percentage points per four weeks. Finance and Information cannot be confidently ordered.
- **Overparameterization**
  - **Method:** Varied random-feature counts above and below training-row counts; compared minimum-norm least squares with ridge.
  - **Result:** Minimum-norm fits interpolate training data but produce extreme future errors. Ridge stabilizes the larger representations; unregularized overparameterization is not competitive.
- **Benchmark scores and nonstationary LLM releases**
  - **Method:** Added ECI/GPQA scores, Poisson regression for a time-varying release rate, and regression for future capability marks; tested arrival predictors and industry interactions.
  - **Result:** Forecast: **12.48 catalog releases in 42 days** (conditional 90% range: 6–20). Benchmark/NHPP boosting MAE: 3.120 versus 3.115 for its control; no clear improvement. Release batching weakens Poisson assumptions.
- **Remaining questions**
  - Do gains persist on future releases? Can official workbooks verify the extract and add predictors?
  - Are rankings and nonlinear expectation terms stable? Does interpolation behavior survive other feature choices?
  - Can benchmark publication histories and release-batch models improve the Poisson analysis? What evidence could identify adoption effects?